

MODEL QUESTION PAPER 2
S4 INSTRUMENTATION ENGINEERING
SUBJECT: MICRO CONTROLLER APPLICATIONS

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	List the sixteen bit registers in 8051	M1.02	R
2	Write the size of Address bus in 8051	M1.01	R
3	List the bits responsible for register bank selection	M1.04	R
4	Name the non maskable interrupt used in 8051	M2.02	R
5	List the power saving modes of 8051	M2.05	R
6	List the instructions used for stack operation	M3.02	R
7	Name the two conditional jump instruction	M3.02	R
8	List classification of stepper motor	M4.07	R
9	Write the command word for increment cursor in LCD	M4.01	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Explain the dual function of port3	M1.01	U
2.	Explain TCON register	M1.04	U
3.	Explain the priority of interrupts	M2.02	U
4.	Explain the stack operation	M2.03	U
5.	Describe the direct and indexed addressing mode with example	M3.01	U
6.	Explain MOVX DPTR @A+PC and MOVC DPTR , @A+PC	M3.02	U
7.	Draw the interfacing diagram of the opto isolator with 8051	M4.06	U
8.	Draw the interfacing diagram of ADC with 8051	M4.03	U
9.	Describe the logical instructions	M3.02	U
10.	Describe PCON register	M1.04	U

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Explain the structure of TMOD Register OR	M1.04	U
IV	Describe the RAM organization of 8051	M1.03	U
V	Explain the modes of operation of timer OR	M2.01	U
VI	Explain the Serial data reception	M2.04	U
VII	Describe data transfer instruction OR	M3.02	U
VIII	Write an ALP to divide two 8 bit numbers and store the result at 9000H	M3.03	A
IX	With necessary diagram explain interfacing of DAC OR	M4.03	U
X	With necessary diagram explain the interfacing of DC motor	M4.07	U
XI	Write an ALP to find the smallest of an array OR	M3.03	A
XII	Write an ALP to find the occurrence of a number	M3.03	A
XIII	With necessary diagram explain the interfacing of LCD OR	M4.01	U

XIV	With necessary diagram explain the interfacing of temperature sensor LM 35	M4.04	U
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Question Wise Analysis

Course : MICRO CONTROLLER APPLICATIONS – QP 2

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.02	R	1	2
I.2	M1.01	R	1	2
I.3	M1.04	R	1	2
I.4	M2.02	R	1	2
I.5	M2.05	R	1	2
I.6	M3.02	R	1	2
I.7	M3.02	R	1	2
I.8	M4.07	R	1	2
I.9	M4.01	R	1	2
II.1	M1.01	U	3	9
II.2	M1.04	U	3	9
II.3	M2.02	U	3	9
II.4	M2.03	U	3	9
II.5	M3.01	U	3	9
II.6	M3.02	U	3	9
II.7	M4.06	U	3	9
II.8	M4.03	U	3	9
II.9	M3.02	U	3	9
II.10	M1.04	U	3	9
III	M1.04	U	7	15
IV	M1.03	U	7	15
V	M2.01	U	7	15
VI	M2.04	U	7	15
VII	M3.02	U	7	15
VIII	M3.03	A	7	15
IX	M4.03	U	7	15
X	M4.07	U	7	15
XI	M3.03	A	7	15
XII	M3.03	A	7	15
XIII	M4.01	U	7	15

XIV	M4.04	U	7	15
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Course: **MICROCONTROLLER APPLICATIONS – QP 2**

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Mark Distribution

Module		Marks\\Module (hi\\ΣHi)*123(±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	14	29	3	3	3	9	2	14	8	26
2	16	33	2	2	2	6	2	14	6	22
3	15	31	2	2	3	9	4	28	9	39
4	15	30	2	2	2	6	4	28	8	36
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	8	6
Understanding	94	76
Applying	21	18
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100